



A.R.T.E.M.I.S

Project Proposal

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Abstract

People often lose valuable time resuming work after interruptions as they re-establish context, relocate files, and recall planned next steps, while simultaneously managing routine manual tasks such as organizing documents, handling emails, and searching for information. Although existing AI assistants can automate some of these tasks, they often rely on constant step-by-step user instruction or demand broad, invasive access to system data. To address these limitations, we present ARTEMIS (Adaptive Reasoning, Task Execution and Multi-agent Intelligence System), a bounded, opt-in desktop AI assistant designed to manage daily workflows within user-defined scopes. ARTEMIS operates strictly on files, notes, and resources explicitly added to a workspace. Powered by agentic AI, the system interprets complex requests, formulates multistep execution plans, and selects appropriate tools to automate tasks like document synthesis, code analysis, email drafting, and web retrieval, with support for voice interaction. To maintain safety and user agency, high impact actions require explicit user approval alongside built-in action previews, logs, and undo options. The system will be evaluated across task completion time, cognitive workload, usability, and user trust to demonstrate that a workspace bounded AI can deliver an effective multi-step automation without compromising user control or data privacy.

1. Introduction

Students, developers, freelancers, and knowledge workers frequently juggle multiple complex tasks across various files, notes, emails, and online resources. When resuming a project after an interruption, users spend substantial time and cognitive effort re-establishing context, recalling prior actions, locating relevant files, and identifying next steps. Moreover, manual administrative chores such as file organization, document synthesis, code analysis, and routine correspondence consume significant daily effort, distracting users from their main primary work.

While existing AI solutions try to alleviate these burdens, existing systems suffer from underlying architectural trade-offs. Traditional systems often require explicit, command by command prompting for every task step, offering minimal proactive assistance. Conversely, modern computer-use AI systems frequently demand broad, unconstrained access to a user's entire operating environment. This lack of clear boundaries creates significant risks regarding data privacy, unintended system modifications, and user mistrust, leaving a critical gap for an assistant that provides multi-step automation within safe boundaries.

To address these limitations we introduce ARTEMIS, a bounded, opt-in desktop AI assistant. ARTEMIS restricts its execution environment to files, notes and context explicitly added by the user to a selected workspace. Using agentic AI, the system interprets user requests, breaks complex tasks into actionable plans, and selects necessary tools to assist with context restoration, document synthesis, code analysis, web search, email preparation, and voice interaction. Rather than acting autonomously, ARTEMIS integrates a human-in-loop approval mechanism that previews major file or network actions before execution. The project will evaluate ARTEMIS across task completion time, cognitive workload, usability, and user trust to determine the efficiency of a bounded assistant in providing effective task automation while preserving user control.

2. Goals and Objectives

Our project's objectives include:

- To develop a bounded AI desktop assistant that helps users with their daily work.
- To help users save time and effort when working with files, documents, code, and routine tasks.
- To provide useful AI assistance while keeping the user in control of important actions
- To improve user trust and privacy by limiting ARTEMIS to a selected workspace
- To evaluate whether ARTEMIS can improve task completion time, usability, and user experience

3. Scope of the Project

The scope of ARTEMIS is to develop a desktop AI assistant that helps users manage their daily work in a selected workspace. ARTEMIS will only access the files, notes, and information that the user gives it. The system will help users continue their previous work, organize files, collect information from documents, automate repeated tasks, understand code, search for information, prepare emails and documents, and use voice interaction.

ARTEMIS will also make sure that the user stays in control. Actions that change files or send information outside the workspace will require user approval before they are performed. The system will also provide previews, activity records, and undo options where needed.

The project will focus on building and evaluating these features within the project timeline. It will not provide full access to the user's computer or perform actions completely on its own. The main focus is to provide useful AI assistance while keeping the system simple, limited, and trustworthy.

4. Initial Study and Work Done so Far

As far as our research is concerned, several studies have already explored the use of AI agents for different tasks. Research has shown that AI agents can understand a task, reason about it, use different tools, and perform multiple steps instead of only giving a simple response [1]. Studies have also explored the use of AI for software development, where AI agents can help find code, generate code, fix errors, and solve programming problems. However, research has also shown that human feedback is important because AI-generated code may not always be correct [2].

Previous studies have also explored the effect of interruptions on computer-based work and how users recover their previous working context after an interruption [3]. Research on Voice User Interfaces has studied voice-based interaction using measures such as task completion time, task success, errors, usability, and user experience [4]. Furthermore, recent research on agentic AI has focused on controlling AI actions, limiting the access given to agents, and involving humans when important actions are performed [5]. These studies helped us understand the current work in this area and identify the limitations of existing approaches.

Our target is to combine these ideas into one bounded desktop AI assistant, ARTEMIS. The system will help users continue their previous work, organize files, work with documents, understand code, perform repeated tasks, search for information, prepare emails and documents, and use voice interaction. ARTEMIS will work only with the workspace selected by the user, and important actions will require user approval. Through this work, we aim to provide useful AI assistance while keeping the user's data, control, and actions protected.

References

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